Lab 2 - 1: The First Observation (Using the AWS Console)

1. Introduction & Scenario

Welcome, Data Manager for "Project Starlight"! Your astronomy club has been working hard, and last night they captured a stunning, high-resolution image of the Andromeda galaxy. It's a fantastic achievement, but now the team is asking a critical question: "Where do we store this data safely?"

Your first task is to create a secure, organized home for this image on Amazon S3. In this lab, you will use the AWS Management Console, a graphical, web-based interface, to accomplish this. Our goal is to build a foundational understanding of S3's core concepts before we move on to automation.

2. Learning Objectives

By the end of this lab, you will be able to:

- Navigate to the S3 service in the AWS Console.
- Create a globally unique S3 bucket.
- Organize data by creating folders (prefixes).
- Upload, view, and delete objects (files).

3. Prerequisites

- An AWS account with IAM user access to the AWS Management Console.
- A sample image file. Please download this image and save it to your computer's Desktop as `andromeda.jpg`.

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4. Lab Steps

Part A: Navigate to the S3 Console

1. Sign in to the **AWS Management Console**.
2. In the main search bar at the top, type S3 and press Enter.
3. Click on the **S3** service result to go to the S3 dashboard. This is your central hub for managing all your storage.

Part B: Create Your First Bucket

An S3 **bucket** is a container for your data. Think of it like a top-level folder or a drive on your computer. Every bucket in all of AWS must have a **globally unique name**.

1. On the S3 dashboard, click the **Create bucket** button.
2. In the **Bucket name** field, you must enter a unique name. We'll use a standard format to guarantee this. Enter a name like: `starlight-data-yourname-` followed by today's date (e.g., `starlight-data-janedoe-20250916`).
3. In the **AWS Region** dropdown, select a region to host your bucket. For this course, we will consistently use **US East (N. Virginia) us-east-1**.

4. Leave all other settings as their defaults for now (like "Block all public access"). These are important security settings that we'll explore later.
5. Scroll to the bottom and click **Create bucket**.

You will be returned to the main dashboard, where you should now see your new bucket in the list. Congratulations, you've created your first piece of cloud infrastructure!

Part C: Create a Folder Structure

Just like on your computer, it's good practice to organize files into folders. In S3, "folders" are actually just organizational placeholders called **prefixes**.

1. Click on your newly created bucket's name to navigate inside it.
2. Click the **Create folder** button.
3. In the **Folder name** field, type `raw-images` and then click **Create folder**.

This simple step is crucial for keeping your project data organized as it grows. You should now see your `raw-images` folder inside the bucket.

Part D: Upload Your First Object

An **object** is the fundamental entity stored in S3—it's simply a file and its metadata. Let's upload the Andromeda galaxy image.

1. Click on the `raw-images` folder to enter it.
2. Click the **Upload** button.
3. On the upload page, you can either drag and drop your file or click the **Add files** button.
4. Click **Add files**, navigate to your Desktop, and select the `andromeda.jpg` file you downloaded earlier.
5. Leave all the default settings and scroll down to click the **Upload** button.

After a moment, you'll see a green banner indicating the upload was successful. You can now click the **Close** button. You should see `andromeda.jpg` listed inside your `raw-images` folder.

Part E: View and Delete the Object

Now that the file is in S3, you can interact with it.

1. Click on the object's name, `andromeda.jpg`. This takes you to the **Object details** page.
2. Here you can see important information, like the **Object URL**, which is the direct address for this file in S3.
3. Let's practice a management action. Navigate back to the `raw-images` folder by clicking the link in the breadcrumb navigation at the top of the page.
4. Select the checkbox next to `andromeda.jpg`.
5. With the file selected, click the **Delete** button that appears near the top of the file list.
6. A confirmation screen will appear. To confirm deletion, you must type `permanently delete` into the text field. Do this now and click the **Delete objects** button.

The object is now removed from your bucket.

Deliverable

To receive completion marks for this lab, you must submit a screenshot demonstrating that you have successfully completed the main task. Your screenshot should clearly show the andromeda.jpg file listed inside the raw-images folder within your S3 bucket.

Part F: Cleanup (Very Important!)

To avoid any unexpected costs and to practice good cloud hygiene, you should always remove resources you are no longer using.

1. Navigate back to the main **S3 buckets** list.
2. Select the checkbox next to the bucket you created (`starlight-data-...`).
3. Click the **Delete** button.
4. A confirmation screen will appear. You must type the full name of your bucket into the text field to confirm. Do this now and click the **Delete bucket** button.

You have successfully completed your first hands-on lab with S3! You've learned how to create storage, organize it, and manage the files within it, all using the AWS Console. In the next lab, we will perform these exact same actions using the command line.